

Exercise 3: Optimality Conditions and Linear Least Squares
(to be returned on November 11th, 8:15)

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The aim of this sheet is to strengthen your knowledge in optimality conditions and least squares estimation.

Exercise Tasks

1. ON PAPER: Given the function $f(x) : \mathbb{R}^n \rightarrow \mathbb{R}$ with $f(x) = x^\top Q x + c^\top x$ and fixed $c \in \mathbb{R}^n$.
 - (a) Consider the not necessarily symmetric matrix $Q \in \mathbb{R}^{n \times n}$ and compute the gradient $\nabla f(x) \in \mathbb{R}^n$ and the Hessian $\nabla^2 f(x) \in \mathbb{R}^{n \times n}$ of this function for any x .
 - (b) If Q is symmetric, what properties does it have to fulfil such that the unique minimizer x^* can be computed?
 - (c) Compute the unique minimizer and the minimum function value $f(x^*)$ under the correct assumptions. (3 points)
2. ON PAPER: We would like to find the parameters $\hat{\theta}_{\text{LS}}$ of a linear model $y(k) = \phi(k)^\top \theta + \epsilon(k)$, where $\epsilon(k) \sim \mathcal{N}(0, \sigma_\epsilon^2)$ is an additive i.i.d. zero-mean Gaussian noise that perturbed a series of N scalar measurements $y_N = [y(1), \dots, y(N)] \in \mathbb{R}^N$. From the lecture we know that $\hat{\theta}_{\text{LS}}$ can be computed using least-squares:

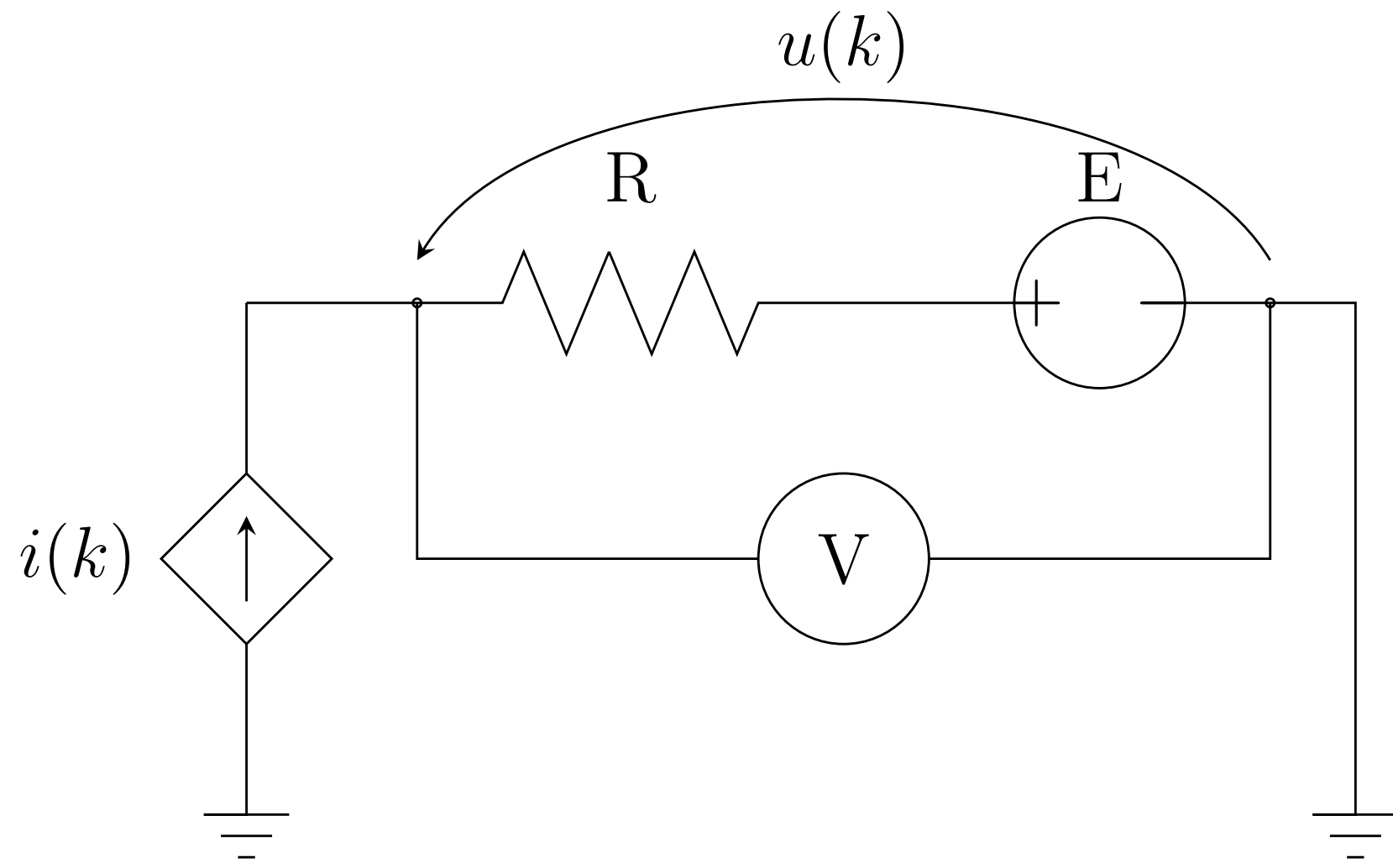
$$\hat{\theta}_{\text{LS}} = \arg \min_{\theta} \frac{1}{2} \|y_N - \Phi_N \theta\|_2^2$$

where $\Phi_N \in \mathbb{R}^{N \times d}$. Assume that σ_ϵ^2 is known.

- (a) State the matrix Φ_N and the solution of least squares problem $\hat{\theta}_{\text{LS}}$.
- (b) Calculate the covariance of the least squares estimate $\text{cov}(\hat{\theta}_{\text{LS}})$.

Hint: Recall from Exercise 2 that the covariance matrix of a vector-valued variable $Y = AX + b$ for a constant $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ is given by $\text{cov}(Y) = A \text{cov}(X) A^\top$. (2 points)

3. Consider the following experimental set up to estimate the values of E and R .



You obtain a dataset containing N measurements of the voltage $u(k)$ for different values of $i(k)$. We assume that the input measurements $i(k)$ is not affected by noise, but that the measurements $u(k)$ are affected by i.i.d. additive noise $n_u(k)$. Under these assumptions the measurement model is given by:

$$u(k) = m(k) + n_u(k) \quad \text{where} \quad m(k) = E + R \cdot i(k).$$

- (a) CODE: Load the dataset containing the measurements. Plot the data with 'x' markers and add labels to both axis. (0.5 points)
- (b) ON PAPER: Formulate the problem of estimating the parameters E and R as a least squares problem where $\theta = [E \ R]^\top \in \mathbb{R}^2$ by defining $\Phi \in \mathbb{R}^{N \times 2}$ and $y \in \mathbb{R}^N$ such that the optimizer is given by $\hat{\theta} = \arg \min_{\theta} \|y - \Phi\theta\|_2^2$. (1 points)
- (c) CODE: Use the least squares estimator formulated in the previous subtask to find the experimental values of R and E for the dataset. Plot the linear fit through the measurement data. (1 points)
- (d) CODE: Plot a histogram of the residuals defined as $r(k) = u(k) - m(k)$, where $m(k) = E + R \cdot i(k)$ is the voltage determined by the model, and $u(k)$ are the obtained measurements. (0.5 points)
- (e) ON PAPER: Give an educated guess for the type of noise distribution. (0.5 points)

As we will learn later in the lecture, for i.i.d. measurement noise, the covariance matrix of the d estimated parameters $\hat{\theta}$ from a single experiment can be approximated by

$$\hat{\Sigma}_{\hat{\theta}} = \frac{\|y - \Phi\hat{\theta}\|_2^2}{N - d} (\Phi^\top \Phi)^{-1}$$

From this estimated covariance matrix, the estimated variances for the single components of the parameter vector can be extracted as the respective diagonal entry.

- (f) CODE: Calculate the estimated variance of both the R and E estimate and print their values along with a 1- σ confidence estimate in the format of

$$(\text{quantity}) = (\text{value}) \pm (1 \text{ standard deviation})$$

(1.5 points)

This sheet gives 10 points in total.

15. $f(x) : \mathbb{R}^n \rightarrow \mathbb{R}$; $f(x) = x^T Q x + c^T x$ and fixed $c \in \mathbb{R}^n$

a). Q is not symmetric but Q is square ($Q \in \mathbb{R}^{n \times n}$)

$$\begin{aligned} \bullet \text{ find } \nabla f(x) &= \frac{\partial}{\partial x} (x^T Q x + c^T x) = \frac{\partial}{\partial x} (x^T Q x) + \frac{\partial}{\partial x} (c^T x) \\ &= \frac{\partial}{\partial x} (x^T Q x) + c^T = (Q + Q^T)x + c \end{aligned}$$

$$\begin{aligned} \bullet \text{ find } \nabla^2 f(x) &= \frac{\partial}{\partial x} [(Q + Q^T)x + c] \\ &= \frac{\partial}{\partial x} [(Q + Q^T)x] + \frac{\partial}{\partial x} c = Q + Q^T \rightarrow \end{aligned}$$

b). Q is symmetric $\rightarrow$ It has definite $\leftarrow$ positive definite
positive semi-definite

c). $\nabla f(x) = 2Qx + c$ | positive definite is important, allowing unique minimum

$$\nabla^2 f(x) = \underbrace{2Q}_{\text{based on "under correct assumption"}}$$

$$\nabla f(x) = 0 \Leftrightarrow 2Qx + c = 0$$

$$\Leftrightarrow Qx = -\frac{1}{2}c$$

$$\Leftrightarrow Q^{-1}Qx = -\frac{1}{2}Q^{-1}c$$

$$\Leftrightarrow \boxed{x^* = -\frac{1}{2}Q^{-1}c} \quad (2)$$

substitute x^* into $f(x) = x^T Q x + c^T x$

$$\Rightarrow f(x^*) = \left(-\frac{1}{2}Q^{-1}c\right)^T \cdot Q \cdot \left(-\frac{1}{2}Q^{-1}c\right) + c^T \cdot \left(-\frac{1}{2}Q^{-1}c\right)$$

$$= -\frac{1}{2}c^T (Q^{-1})^T \cdot Q \cdot \left(-\frac{1}{2}\right)Q^{-1}c - \frac{1}{2}c^T Q^{-1}c$$

$$= +\frac{1}{4}c^T \underbrace{Q^{-1}Q}_{\pm} Q^{-1}c - \frac{1}{2}c^T Q^{-1}c$$

$$= \frac{1}{4}c^T Q^{-1}c - \frac{1}{2}c^T Q^{-1}c = \boxed{-\frac{1}{4}c^T Q^{-1}c} \rightarrow \text{optimal value of } f(x) \text{ with optimized } x^*.$$

2. $\hat{\theta}_{LS}$ of $y(k) = \varphi(k)^T \theta + \varepsilon(k)$, $\varepsilon(k) \sim \mathcal{N}(0, \sigma_\varepsilon^2)$
Noise

a) $\Phi_N \in \mathbb{R}^{N \times d}$: regressor matrix.

- N : represents one complete observation or measurement in time.
- d : Every column represents a specific input feature or variable that corresponds to one of the unknown parameters in vector θ .

→ for example: $L(k) = \theta_1 T(k) + \theta_2 \cdot 1$

• vector $\theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$

• regressor matrix: $\Phi_N = \begin{bmatrix} T(k_1) & 1 \\ T(k_2) & 1 \\ \vdots & \vdots \\ T(k_N) & 1 \end{bmatrix} \rightarrow N \times d$

$$\Phi_N = \begin{bmatrix} \phi^T(1) \\ \phi^T(2) \\ \phi^T(3) \\ \vdots \\ \phi^T(N) \end{bmatrix}$$

$$\text{or } [\phi(1) \ \phi(2) \ \dots \ \phi(N)]^T$$

note: $[\phi^T(1) \ \phi^T(2) \ \dots \ \phi^T(N)]$ is incorrect

because: $\phi^T(1)$ is $1 \times d$ vector.

$$\rightarrow [\phi^T(1) \ \phi^T(2) \ \dots \ \phi^T(N)]^T$$

which result $\begin{bmatrix} \phi(1) \\ \phi(2) \\ \vdots \\ \phi(N) \end{bmatrix} \rightarrow N(d \times 1) \text{ matrix}$

but we need $N \times d$.

$$\rightarrow [\phi(1) \ \phi(2) \ \dots \ \phi(N)]^T$$

$$= \begin{bmatrix} \phi^T(1) \\ \phi^T(2) \\ \vdots \\ \phi^T(N) \end{bmatrix} \Rightarrow N \times d \text{ matrix.}$$

$\phi = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$
 $\rightarrow \theta^T = [\theta_1 \ \theta_2]$
 $\underbrace{[\theta_1 \ \theta_2]}_1 \underbrace{[\theta_1 \ \theta_2]}_2^T = \begin{bmatrix} \theta_1 & \theta_1 \\ \theta_2 & \theta_2 \end{bmatrix} \rightarrow 2(d \times 1)$

Correct intuition.

$$\underbrace{\begin{bmatrix} \theta_1 & \theta_1 \\ \theta_2 & \theta_2 \end{bmatrix}}_1^T = \underbrace{\begin{bmatrix} \theta_1 & \theta_2 \\ \theta_1 & \theta_2 \end{bmatrix}}_2 (N \times d)$$

$$\hat{\theta}_{LS} = \underset{\theta}{\operatorname{argmin}} \frac{1}{2} \|y_N - \Phi_N \theta\|_2^2$$

$J(\theta)$: cost function or objective function

$$f(\theta) = \frac{1}{2} \|y_N - \Phi_N \theta\|_2^2$$

reminder: L_2 -Norm of $x = [x_1, \dots, x_n]$

$$\Rightarrow \|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

$$\text{from that} \Rightarrow \|x\|_2^2 = x_1^2 + x_2^2 + \dots + x_n^2$$

$f(\theta)$ can be rewritten as

$$f(\theta) = \frac{1}{2} (y_N - \Phi_N \theta)^T (y_N - \Phi_N \theta)$$

$$= \frac{1}{2} (y_N^T - (\Phi_N \theta)^T) (y_N - \Phi_N \theta) = \frac{1}{2} (y_N^T - \theta^T \Phi_N^T) (y_N - \Phi_N \theta)$$

$$= \frac{1}{2} (y_N^T y_N - \underbrace{y_N^T \Phi_N \theta}_{\text{scalar}} - \underbrace{\theta^T \Phi_N^T y_N}_{\text{scalar}} + \theta^T \Phi_N^T \Phi_N \theta)$$

=

Identical

$$= \frac{1}{2} (y_N^T y_N - 2 y_N^T \Phi_N \theta + \theta^T \Phi_N^T \Phi_N \theta)$$

→ gradient of $f(\theta)$

$$\nabla f(\theta) = \frac{\partial}{\partial \theta} \left[\frac{1}{2} (y_N^T y_N - 2 y_N^T \Phi_N \theta + \theta^T \Phi_N^T \Phi_N \theta) \right]$$

0

Vector

c. x

Q : symmetric matrix
 $x^T Q x$

$$= \frac{1}{2} (-2 (y_N^T \Phi_N)^T + 2 \Phi_N^T \Phi_N \cdot \theta)$$

$$= \Phi_N^T \Phi_N \theta - \Phi_N^T y_N$$

Let gradient $\nabla f(\theta) = 0$

$$\Leftrightarrow \Phi_N^T \Phi_N \theta - \Phi_N^T y_N = 0 \Leftrightarrow \Phi_N^T \Phi_N \theta = \Phi_N^T y_N$$

$$\Leftrightarrow \theta^* = (\Phi_N^T \Phi_N)^{-1} \Phi_N^T y_N$$

Φ_N is a rectangular matrix → inverse in standard form.

If Φ is square matrix → can do the following up.

$$\Leftrightarrow \theta^* = \Phi_N^{-1} (\Phi_N^T)^{-1} \Phi_N^T y_N = \Phi_N^{-1} \cdot y_N$$

→ optimized solution.

$$2b) \text{cov}(\hat{\theta}_{LS}) = ?$$

$$\hat{\theta}_{LS} = \underbrace{\left[(\Phi_N^T \Phi_N)^{-1} \Phi_N^T \right]}_A y_N \quad X$$

$$Y = AX + b$$

$$\text{cov}(Y) = A \text{cov}(X) A^T$$

$$\Rightarrow \text{cov}(\hat{\theta}_{LS}) = A \text{cov}(y_N) A^T$$

where $\text{cov}(y_N)$ is σ_e^2 because with $y(k) = \phi(k)^T \theta + \epsilon(k)$

• $\phi(k)^T \theta$ is constant
 • $\epsilon(k)$ is the noise
 } the randomness is completely comes from the noise $\epsilon(k)$

$$\Rightarrow \text{the covariance matrix } \text{cov}(y_N) = \begin{bmatrix} \sigma_e^2 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \sigma_e^2 \end{bmatrix}$$

$$= \sigma_e^2 I \quad (I \in \mathbb{R}^{n \times n})$$

$$\Rightarrow \text{cov}(\hat{\theta}_{LS}) = \left[(\Phi_N^T \Phi_N)^{-1} \Phi_N^T \right] \sigma_e^2 I \left[(\Phi_N^T \Phi_N)^{-1} \Phi_N^T \right]^T$$

$$= \sigma_e^2 \left[\underbrace{(\Phi_N^T \Phi_N)^{-1}}_{Q^{-1}} \underbrace{(\Phi_N^T \Phi_N)}_Q \right]$$

$$= \sigma_e^2 \cdot \left[I_N \right]^T = \sigma_e^2 (I_N - \Phi_N^T \Phi_N)^{-1}$$

3). b)

$$u(k) = m(k) + n_u(k) \quad \text{where } m(k) = F + R i(k)$$

$$\Rightarrow \underbrace{u(k)}_{\text{measured}} = \underbrace{F + R i(k)}_{\text{measured}} + \underbrace{n_u(k)}_{\text{noise}}$$

$$\Rightarrow \text{Parameters for optimization: } \theta = \begin{bmatrix} F \\ R \end{bmatrix}$$

define Φ_N and y_N for the optimizer: $\hat{\theta} = \arg \min_{\theta} \|y - \Phi \theta\|_2^2$

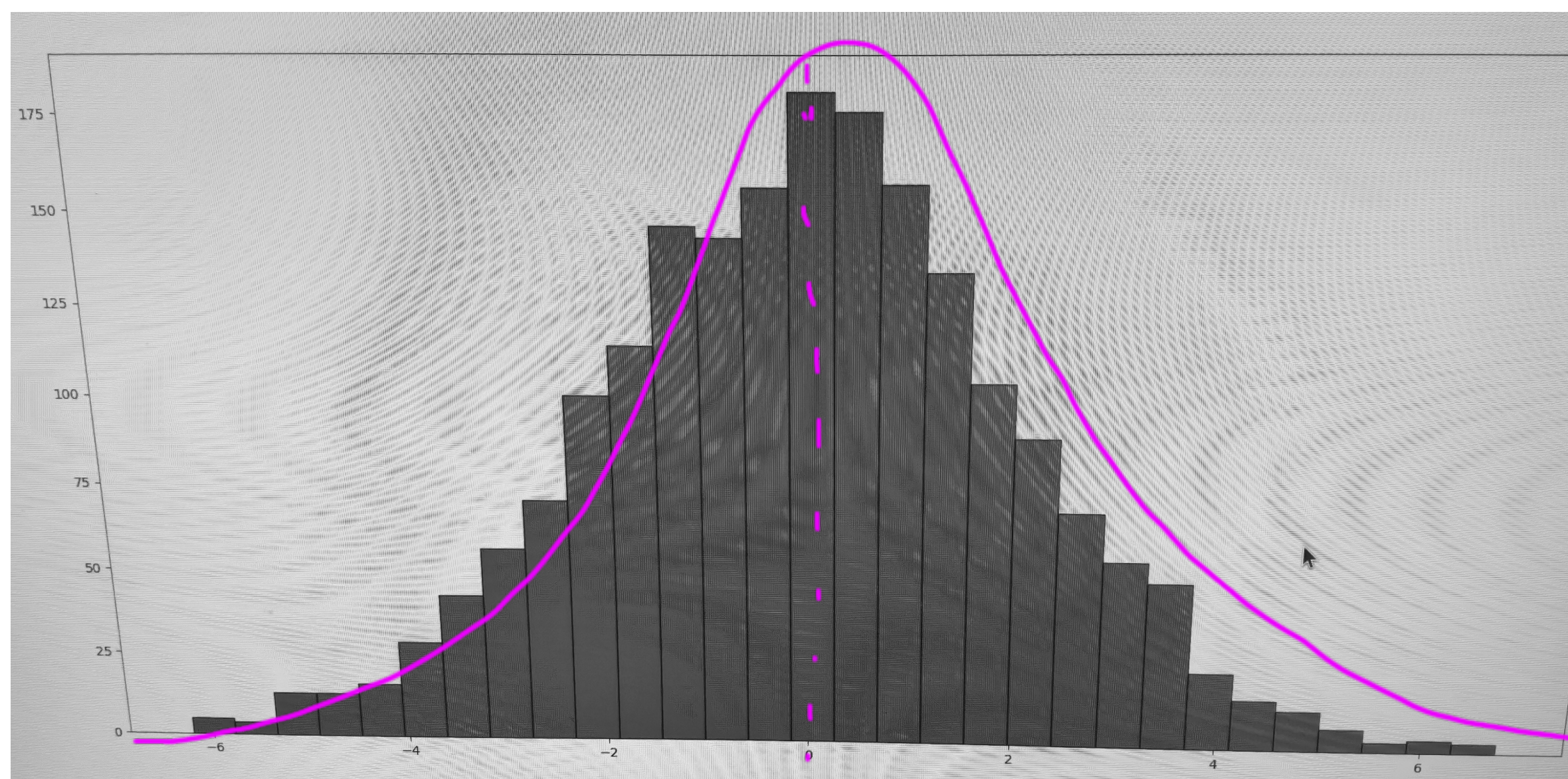
$$\Phi_N = \begin{bmatrix} 1 & i(1) \\ 1 & i(2) \\ \vdots & \vdots \\ 1 & i(n) \end{bmatrix} \in \mathbb{R}^{n \times 2}$$

$$y_N = [u(1) \quad u(2) \quad \dots \quad u(n)]^T$$

Note: the linear equation $y_N = \Phi_N \cdot \theta$ has 3 possibilities:

- variables > measurements: no exact solution $\rightarrow$ infinite analytical solutions
- variables = measurements: one exact solution (noiseless measurement required)
- variables < measurements:

3e).



• since the residual of measurement $x(k)$ is the experimental estimation of the noise $n_u(k)$

$$\rightarrow n_u \sim \mathcal{N}(0, \Sigma_{n_u})$$

• from the graph:

- most of the residuals are located at the center (0)

- the shape of the residuals is look like a bell curve

$\rightarrow$ Noise is followed the Gaussian Dist.

3f. Result in code:

$$\hat{\theta} = \begin{bmatrix} E \\ \mu \end{bmatrix} = \begin{bmatrix} 4.939 \dots \\ 2027.476 \dots \end{bmatrix} \pm \begin{bmatrix} 0.001744 \dots \\ 15.88644 \dots \end{bmatrix}$$

mean
standard deviation.